```
matlab
clear;
clc;
close all;

% Design Parameters
Fs = 5000;
T = 1/Fs;
fc = 1000;
Omega_c = 2*pi*fc;

% Analog Filter Prototype
b_a = Omega_c;
a_a = [1 Omega_c];

% ========================================
% 1. IMPULSE INVARIANCE METHOD
% ========================================
% Analog transfer function:
% H(s) = Omega_c / (s + Omega_c)

p = -Omega_c;
residue_value = Omega_c;

% Impulse invariance conversion
a_imp = [1 -exp(p*T)];
b_imp = [0 residue_value*T];

% ========================================
% 2. BILINEAR TRANSFORMATION METHOD
% ========================================
% Prewarping the cutoff frequency
Omega_p = (2/T)*tan(Omega_c*T/2);

% Bilinear transformation coefficients
K = 2/T;

b_blt = [Omega_p Omega_p]/(K + Omega_p);
a_blt = [1 (Omega_p-K)/(Omega_p+K)];

% ========================================
% DISPLAY NUMERICAL OUTPUTS
% ========================================
fprintf('EXPERIMENT 3: IMPULSE INVARIANCE vs BILINEAR TRANSFORMATION\n\n');

fprintf('Analog Cutoff Frequency: %.2f rad/s\n',Omega_c);
fprintf('Prewarped Analog Frequency: %.2f rad/s\n\n',Omega_p);

fprintf('IMPULSE INVARIANCE COEFFICIENTS\n');
fprintf('Numerator:\n');
disp(b_imp);
fprintf('Denominator:\n');
```

```
disp(a_imp);

fprintf('BILINEAR TRANSFORMATION COEFFICIENTS\n');
fprintf('Numerator:\n');
disp(b_blt);
fprintf('Denominator:\n');
disp(a_blt);

% ========================================
% FREQUENCY RESPONSE CALCULATION
% ========================================
f = linspace(0,Fs/2,1024);
w = 2*pi*f/Fs;

z_inverse = exp(-1i*w);

H_imp = (b_imp(1) + b_imp(2)*z_inverse) ./ ...
        (a_imp(1) + a_imp(2)*z_inverse);

H_blt = (b_blt(1) + b_blt(2)*z_inverse) ./ ...
        (a_blt(1) + a_blt(2)*z_inverse);

% ========================================
% PLOTTING
% ========================================
figure('Name','Impulse Invariance vs Bilinear Transformation');

subplot(2,1,1);

plot(f,abs(H_imp),'b','LineWidth',1.5);
hold on;
plot(f,abs(H_blt),'r--','LineWidth',1.5);
grid on;

legend('Impulse Invariance','Bilinear Transformation');
title('Magnitude Response Comparison');
xlabel('Frequency (Hz)');
ylabel('Absolute Magnitude');

subplot(2,1,2);

plot(f,angle(H_imp)*180/pi,'b','LineWidth',1.5);
hold on;
plot(f,angle(H_blt)*180/pi,'r--','LineWidth',1.5);
grid on;

legend('Impulse Invariance','Bilinear Transformation');
title('Phase Response Comparison');
xlabel('Frequency (Hz)');
ylabel('Phase (degrees)');
```

EXPERIMENT 3: IMPULSE INVARIANCE vs BILINEAR TRANSFORMATION

Analog Cutoff Frequency: 6283.19 rad/s

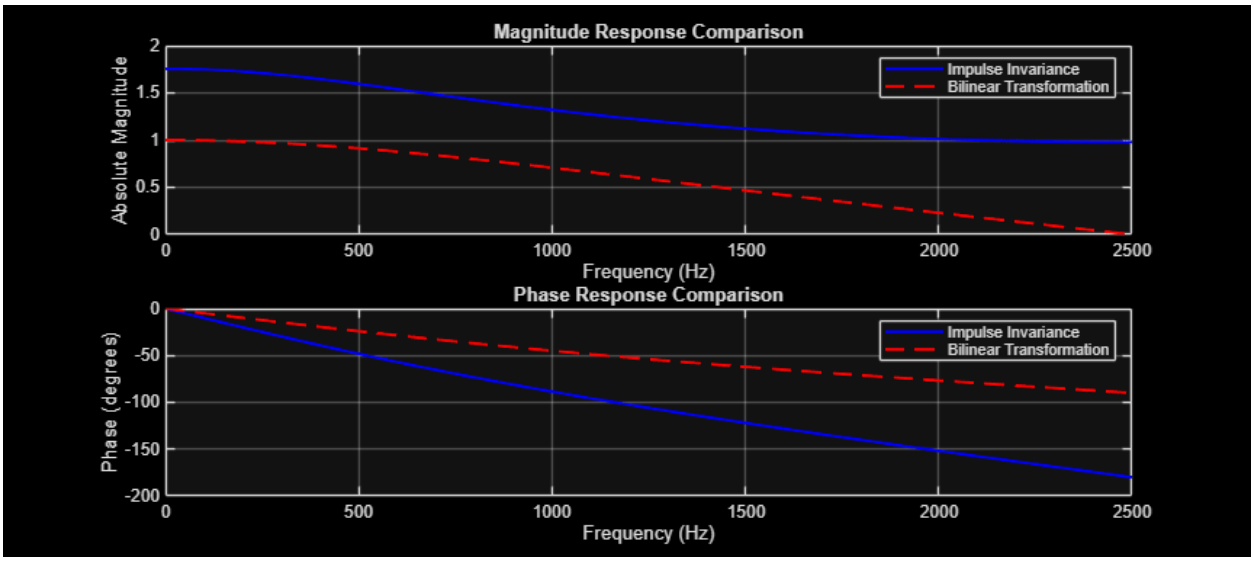
```
Prewarped Analog Frequency: 7265.43 rad/s

IMPULSE INVARIANCE COEFFICIENTS
Numerator:
         0    1.2566

Denominator:
    1.0000   -0.2846

BILINEAR TRANSFORMATION COEFFICIENTS
Numerator:
    0.4208    0.4208

Denominator:
    1.0000   -0.1584
```



Published with MATLAB® R2025b